

L8: Dynamic Communities – Approach #1

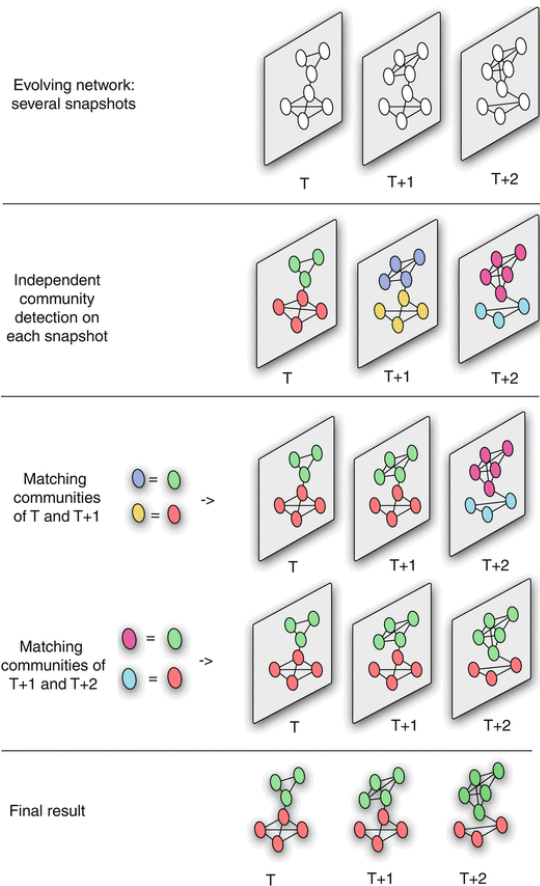


Figure 2 from [Dynamic Community Detection](https://doi.org/10.1007/978-1-4614-6170-8_383) (https://doi.org/10.1007/978-1-4614-6170-8_383) by Cazabet R., Amblard F.

The first approach is that a community detection algorithm is applied independently on each network snapshot, without any constraints imposed by earlier or later network snapshots.

Then, the communities identified at time $T+1$ are matched to the communities identified at time T based on maximum node and edge similarity.

This process is illustrated in the visualization for three successive snapshots. Note that the green community experiences gradual growth while the red community experiences contraction.

The main advantage of this approach is its simplicity, given that we can apply any algorithm for community detection on static networks.

Its main disadvantage however is that the communities of successive snapshots can be significantly different, not because there is a genuine change in the community structure, but because the static community detection algorithm may be unstable, producing very different results even with minor changes in the topology of the input graph.